

1. Using the symbolic representations

b: I ride a bicycle to school.

s: It is snowing.

r: It is raining.

express the following statements in symbolic form.

- a. It is raining and I rode my bicycle to school.

$$r \wedge b$$

- b. If it is raining or snowing, then I will not ride my bicycle to school.

$$(r \vee s) \rightarrow \sim b$$

2. Using the statement, "If your order is over \$50, then you will not have to pay shipping fees." and parts (a), (b), and (c) below, complete the following:

- a. If you do not have to pay shipping fees then your order is over \$50.
- $\sim p \rightarrow o$

- b. If your order is not over \$50 then you have to pay shipping fees.
- $\sim o \rightarrow p$

- c. If you have to pay shipping fees then your order is not over \$50.
- $p \rightarrow \sim o$

Now write the statement as:  $o \rightarrow \sim p$ Identify the converse of the statement. The converse is  $\sim p \rightarrow o$  (a)Identify the inverse of the statement. The inverse is  $\sim o \rightarrow \sim (\sim p)$  (b)Identify the contrapositive of the statement. The contrapositive is  $p \rightarrow \sim o$  (c)

3. For parts 3a-3d, identify the negation of the provided statement.

- a. Some students are taking Math 107.

(a) All students are taking Math 107.

(b) No students are taking Math 107.

(c) Some students are not taking Math 107.

(d) Some students are taking Math 107.

negate

|      |                       |      |
|------|-----------------------|------|
| all  | $\longleftrightarrow$ | some |
| none | $\leftrightarrow$     |      |

- b. All players need to attend practice.

(a) All players need to attend practice.

(b) No players need to attend practice.

(c) Some players do not need to attend practice.

(d) Some players need to attend practice.

- c. No professors know how to read.

(a) All professors know how to read.

(b) No professors know how to read.

(c) Some professors do not know how to read.

(d) Some professors know how to read.

- d. Some students are going to lunch.

(a) All students are going to lunch.

(b) No students are going to lunch.

(c) Some students are not going to lunch.

(d) Some students are going to lunch.

4. Use a truth table to determine if the following arguments are valid.

- a. b: You buy the bike.

t: You buy the tent.

1. You buy the bike or the tent.

2. You buy the bike.

Therefore, you do not buy the tent.

|   |   | 1          | 2 | c            |          | Argument                     |
|---|---|------------|---|--------------|----------|------------------------------|
| b | t | $b \vee t$ | b | $1 \wedge 2$ | $\sim t$ | $(1 \wedge 2) \rightarrow c$ |
| T | T | T          | T | T            | F        | F                            |
| T | F | T          | T | T            | T        | T                            |
| F | T | T          | F | F            | F        | T                            |
| F | F | F          | F | F            | T        | T                            |

The argument is not always true so it is, **invalid**

- b.  $r$ : You register early. 1. If you register early, then you will receive a discount.  
 $d$ : You receive a discount. 2. You register early.  
 -----  
 Therefore, you will receive a discount.

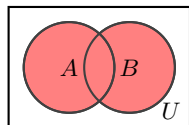
|     |     | 1                 | 2   |              | c   | Argument                     |
|-----|-----|-------------------|-----|--------------|-----|------------------------------|
| $r$ | $d$ | $r \rightarrow d$ | $r$ | $1 \wedge 2$ | $r$ | $(1 \wedge 2) \rightarrow c$ |
| T   | T   | T                 | T   | T            | T   | T                            |
| T   | F   | F                 | T   | F            | T   | T                            |
| F   | T   | T                 | F   | F            | F   | T                            |
| F   | F   | T                 | F   | F            | F   | T                            |

The argument is always true so it is, **valid**

5. Let  $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$  be the universal set, and  $A = \{1, 2, 3, 4\}$  and  $B = \{4, 6, 8\}$  be subsets of  $U$ . Find the indicated set, and draw a Venn diagram corresponding to the indicated set.

- a.  $A \cup B$

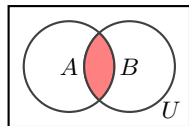
$$A \cup B = \{1, 2, 3, 4, 6, 8\}$$



The *Union*: elements in either  $A$  or  $B$ . Note that  $A, B \subseteq A \cup B$ . Do not repeat elements.

- b.  $A \cap B$

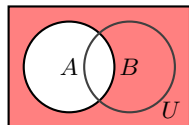
$$A \cap B = \{4\}$$



The *intersection*: elements in both  $A$  and  $B$ . Note that  $A \cap B \subseteq A, B$ . The intersection can be the empty set,  $\emptyset$ , e.g.,  $A \cap A' = \emptyset$ .

- c.  $A'$

$$A' = \{5, 6, 7, 8\}$$

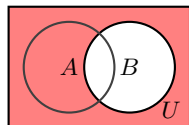


The *complement* of  $A$ : elements in  $U$  but not in  $A$ . Note that  $A' \cup A = U$ . In terms of set difference,

$$\begin{aligned} A' &= U - A \\ &= \{\cancel{1}, \cancel{2}, \cancel{3}, \cancel{4}, 5, 6, 7, 8\} = \{5, 6, 7, 8\} \end{aligned}$$

- d.  $B'$

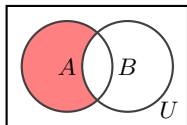
$$B' = \{1, 2, 3, 5, 7\}$$



$$\begin{aligned} B' &= U - B \\ &= \{1, 2, 3, \cancel{4}, 5, \cancel{6}, 7, \cancel{8}\} = \{1, 2, 3, 5, 7\} \end{aligned}$$

e.  $A \cap B'$ 

$$A \cap B' = \{1, 2, 3\}$$



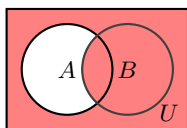
The intersection is just what both sets have in common,

$$\begin{aligned} B' &= \{1, 2, 3, 5, 7\} \\ A &= \{1, 2, 3, 4\} \\ A \cap B' &= \{1, 2, 3\} \end{aligned}$$

Equivalently, this is everything in  $A$  but not in  $B$ , so  $A \cap B' = A - B$ .

f.  $A' \cup B$ 

$$A' \cup B = \{4, 5, 6, 7, 8\}$$

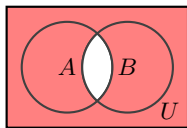


The union is just everything in both sets; don't repeat elements.

$$\begin{aligned} A' &= \{5, 6, 7, 8\} \\ B &= \{4, 6, 8\} \\ A' \cup B &= \{5, 6, 7, 8, 4, \cancel{6}, \cancel{8}\} = \{4, 5, 6, 7, 8\} \end{aligned}$$

g.  $(A \cap B)'$ 

$$(A \cap B)' = \{1, 2, 3, 5, 6, 7, 8\}$$

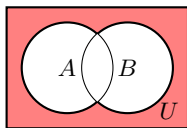


First find  $A \cap B$ . From above,  $A \cap B = \{4\}$

$$\begin{aligned} (A \cap B)' &= U - (A \cap B) \\ &= \{1, 2, 3, \cancel{4}, 5, 6, 7, 8\} = \{1, 2, 3, 5, 6, 7, 8\} \end{aligned}$$

h.  $(A \cup B)'$ 

$$(A \cup B)' = \{5, 7\}$$



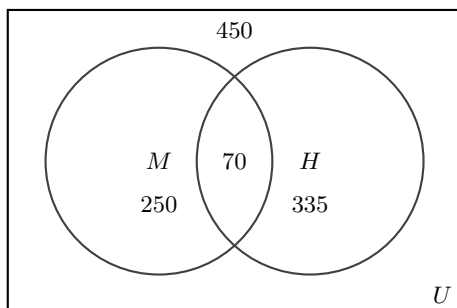
First find  $A \cup B$ . From above  $A \cup B = \{1, 2, 3, 4, 6, 8\}$ .

$$\begin{aligned} (A \cup B)' &= U - (A \cup B) \\ &= \{\cancel{1}, \cancel{2}, \cancel{3}, \cancel{4}, 5, \cancel{6}, 7, \cancel{8}\} = \{5, 7\} \end{aligned}$$

6. A college has 1105 students. This semester 320 students are taking a math course, 405 of the students are taking a history class, and 250 of the students taking a math class are not enrolled in a history class.

- Construct a Venn diagram, and write the appropriate numbers in the four regions
- How many of the students are taking math and history?
- How many students are taking history but not math?
- How many students are taking math or history?
- How many students are taking neither math nor history?

(a)  $n(M \cap H) = 320 - 250 = 70$ ,  $n(H - M) = 405 - 70 = 335$ , and  $n((M \cup H)') = 1105 - 250 - 70 - 335 = 450$ .

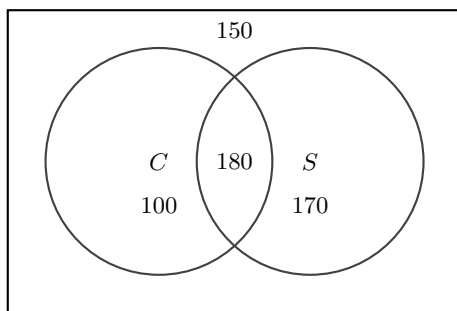


- $n(H \cap M) = 70$
- $n(H - M) = 335$
- $n(M \cup H) = 250 + 70 + 335 = 655$
- $n((M \cup H)') = n(U) - n(M \cup H) = 1105 - 655 = 450$

7. *Survivor* and *CSI* belong to the Thursday night prime time lineup. A group of 600 college students were surveyed. 350 students watch *Survivor* and 280 watch *CSI*. 180 students watch both shows.

- Construct a Venn diagram, and write the appropriate numbers in the eight regions.
- How many students watch only *Survivor*?
- How many students watch *CSI* or *Survivor*?
- How many students watch only *CSI*?

(a)  $n(C \cap S) = 180$ ,  $n(C - S) = 280 - 180 = 100$ ,  $n(S - C) = 350 - 180 = 170$ , and  $n((C \cup S)') = 600 - (180 + 170 + 100) = 150$



- From diagram: 170
- From diagram:  $n(C \cup S) = 100 + 170 + 180 = 450$
- From diagram: 100

8. Find the value of each of the following:

- $4!$

$$4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$$

b.  $7!$ 

$$7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5040$$

c.  $0!$ 

$$0! = 1$$

d.  $\frac{10!}{7!}$ 

$$\frac{10!}{7!} = \frac{10 \cdot 9 \cdot 8 \cdot \cancel{7} \cdot \cancel{6} \cdot \cancel{5} \cdot \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot \cancel{1}}{\cancel{7} \cdot \cancel{6} \cdot \cancel{5} \cdot \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot \cancel{1}} = 720$$

e.  $\frac{27!}{4!25!}$ 

$$\frac{27!}{4!25!} = \frac{27 \cdot 26}{4! \cdot 1} = \frac{702}{24} = 117. \text{ Similarly to above, everything before 25 cancels out.}$$

f.  $\frac{1000!}{998!}$ 

$$\frac{1000!}{998!} = \frac{1000 \cdot 999}{1} = 999,000. \text{ Similarly to above, everything before 999 cancels out.}$$

g.  ${}_7P_2$ 

$${}_7P_2 = \frac{7!}{(7-2)!} = \frac{7!}{5!} = \frac{5040}{120} = 42 \text{ or } \frac{7!}{5!} = 7 \cdot 6 = 42.$$

h.  ${}_6C_4$ 

$${}_6C_4 = \frac{6!}{(6-4)!4!} = \frac{6!}{2!4!} = \frac{6 \cdot 5}{2!} = \frac{30}{2} = 15$$

9. A certain model of pickup truck is available in five exterior colors, three interior colors, and three interior styles. In addition, the transmission can be either manual or automatic, and the truck can have either two-wheel or four-wheel drive. How many different versions of the pickup truck can be ordered?

We have five categories,  $\_ \times \_ \times \_ \times \_ \times \_$ . Filling in the number of choices for each category we have,

$$\underline{5} \times \underline{3} \times \underline{3} \times \underline{2} \times \underline{2} = 180$$

10. A certain fraternity must elect a president, vice president, and treasurer from its 30 members. How many ways can this be done?

There are no repeats and order does matter (since order will determine who is pres., vice pres., and treasurer). So we use permutations,

$${}_{30}P_3 = \frac{30!}{(30-3)!} = \frac{30!}{27!} = 30 \cdot 29 \cdot 28 = 24,360$$

11. A three-person committee must be chosen from a group of ten men and twelve women. How many committees are possible if it must consist of the following?

- Three men.
- Three women.
- One woman and two men.
- One man and two women.
- One man and two women, and one of these is selected as a spokesperson of the committee.

$$(a) \text{ No repeats and order doesn't matter} \Rightarrow \text{combination. } {}_{10}C_3 = \frac{10!}{(10-3)!3!} = \frac{10 \cdot 9 \cdot 8}{3!} = \frac{720}{6} = 120$$

$$(b) \text{ No repeats and order doesn't matter} \Rightarrow \text{combination. } {}_{12}C_3 = \frac{12!}{(12-3)!3!} = \frac{12 \cdot 11 \cdot 10}{3!} = \frac{1320}{6} = 220$$

- (c) Here we have two categories:  $\_ \times \_$ . For the first categories we have 12 choices (equivalently,  ${}_{12}C_1 = \frac{12!}{(12-1)!1!} = 12$ ), and for the second we have  ${}_{10}C_2 = \frac{10!}{(10-2)!2!} = \frac{10 \cdot 9}{2} = 45$ . So we have  $12 \times 45 = 540$  possibilities.
- (d) Similarly to above, we have two categories. For the first categories we have 10 choices, and for the second we have  ${}_{12}C_2 = \frac{12!}{(12-2)!2!} = \frac{12 \cdot 11}{2} = 66$ . Then we select one of the three to be a spokesperson. So we have  $10 \times 66 \times 3 = 1980$  possibilities.
- Equivalently, use a permutation where the order of 2 members does not matter,  $(10 \times \frac{12!}{10!} \times 3)/2! = 1980$  (like counting the ways to arrange the letters in "SCHOOL"; the order of the "O"'s does not matter).

12. A card is dealt from a well-shuffled, standard deck of 52 cards. Find the probability of being dealt each of the following.

a. A red card

$$\frac{n(\text{red cards})}{n(\text{cards})} = \frac{26}{52} = \frac{1}{2}$$

b. A queen

$$\frac{n(\text{queens})}{n(\text{cards})} = \frac{4}{52} = \frac{1}{13}$$

c. A club

$$\frac{n(\text{clubs})}{n(\text{cards})} = \frac{13}{52} = \frac{1}{4}$$

d. The queen of clubs

$$\frac{n(\text{queen of clubs})}{n(\text{cards})} = \frac{1}{52}$$

e. A queen or a club

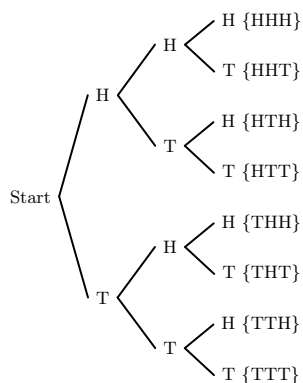
$$\frac{n(\text{queens or clubs})}{n(\text{cards})} = \frac{4 + 13 - 1}{52} = \frac{16}{52} = \frac{4}{13}$$

f. not a queen

$$1 - \frac{4}{52} = \frac{48}{52} = \frac{12}{13}$$

13. Three coins are tossed. Find each of the following:

a. the sample space



b. the event,  $E_1$ , that exactly two are tails

$$E_1 = \{HTT, THT, TTH\}$$

c. the event,  $E_2$ , that at least two are tails.

$$E_2 = \{HTT, THT, TTH, TTT\}$$

d. the probability of  $E_1$

$$p(E_1) = \frac{3}{8}$$

e. the probability of  $E_2$

$$p(E_2) = \frac{4}{8} = \frac{1}{2}$$

14. A drawer has 7 socks. 4 socks are black and 3 are white socks. John randomly pulls out 4 socks. Find the probability of the following:

a. all 4 socks are black.

$$\frac{{}_4C_4}{{}_7C_4} = \frac{1}{35}$$

b. exactly 2 are white.

$$\frac{{}_3C_2 \cdot {}_4C_2}{{}_7C_4} = \frac{3 \cdot 6}{35} = \frac{18}{35}$$

2 are white and 2 are black

c. at least 3 are white.

$$\frac{{}_3C_3 \cdot {}_4C_0 + {}_3C_4 \cdot {}_4C_0}{{}_7C_4} = \frac{1 \cdot 1 + 0 \cdot 1}{35} = \frac{1}{35}$$

3 OR 4 are white

15. Of all the flashlights in a large shipment, 15% have a defective bulb, 10% have a defective battery, and 5% have both defects. If you buy one of these flashlights, find the probability of the following:

- a. a defective bulb or a defective battery

$$p(Batt' \cup Bulb') = .10 + .05 + .05 = .20$$

- b. a good bulb or a good battery

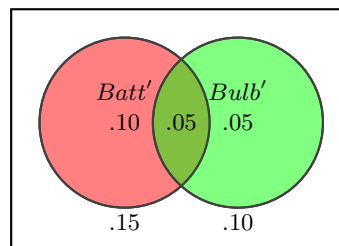
$$p((Batt' \cap Bulb')) = 1 - .05 = .95$$

- c. a defective bulb and a defective battery.

$$p(Bulb' \cap Batt') = .05$$

- d. just a defective bulb

$$p(Bulb' - Batt') = .10 - .05 = .05$$



16. Use the provided table to answer the following questions.

| Age   | Obama | McCain | Other/NA |
|-------|-------|--------|----------|
| 18-29 | 11.9% | 5.8%   | 0.4%     |
| 30-44 | 15.1% | 13.3%  | 0.6%     |
| 45-64 | 18.5% | 18.1%  | 0.4%     |
| 65+   | 7.2%  | 8.5%   | 0.3%     |

- a. What is the probability that a voter under 45 voted for McCain.

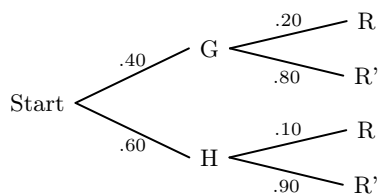
$$p(M | \text{under } 45) = \frac{n(M \cap \text{under } 45)}{n(\text{under } 45)} = \frac{.058 + .133}{.119 + .058 + .004 + .151 + .133 + .006} = \frac{.191}{.471} \approx 40.55\%$$

- b. What is the probability that a person who voted for McCain is 45 or older.

$$p(\text{over } 45 | M) = \frac{n(\text{over } 45 \cap M)}{n(M)} = \frac{.181 + .085}{.058 + .133 + .181 + .085} = \frac{.266}{.457} \approx 58.21\%$$

17. Gregor's Garden Corner buys 40% of its plants from the Green Growery and the rest from Herb's Herbs. Of the plants from the Green Growery, 20% must be returned, and 10% of those from Herb's Herbs must be returned.

- a. draw a tree diagram describing the probability



- b. find the probability that a plant must be returned and it is from Herb's Herbs.

$$p(H \cap R) = (.60)(.10) = .06$$

- c. find the probability that a plant from the Green Growery must be returned.

$$p(R | G) = .20$$

- d. find the probability that a plant must be returned.

A returned plant can be from Herb's OR Gregory,

$$p(R) = p(H \cap R) + p(G \cap R) = (.6)(.1) + (.4)(.2) = .06 + .08 = .14$$

18. Forty randomly selected families were surveyed to determine the number of children in each family. The following results were found:

3.7 3.6 4.0 3.8 4.1 3.9 3.5 3.9  
3.6 4.0 3.6 3.7 3.9 3.3 3.6 4.0

- a. Find the mean.

$$\bar{x} = \frac{\sum x}{n} = \frac{60.2}{16} \approx 3.7625$$

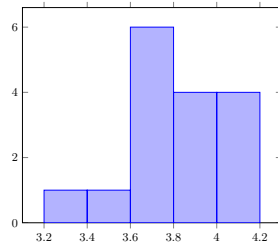
- b. Find the median.

$$\frac{3.7 + 3.8}{2} = 3.75$$

c. Find the standard deviation.

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}} \approx .2217$$

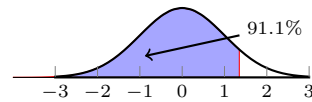
d. Construct a histogram using 5 subintervals, beginning at 3.2, and ending at 4.2.



19. Find the following probabilities,

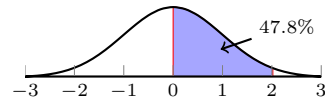
a.  $p(z < 1.35)$

$$p(z < 1.35) \approx 91.15\%$$



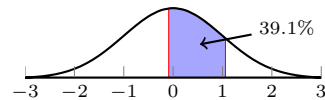
b.  $p(0 < z < 2.02)$

$$p(0 < z < 2.02) \approx 47.83\%$$



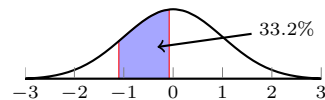
c.  $p(-.09 < z < 1.06)$

$$p(-.09 < z < 1.06) \approx 39.13\%$$



d.  $p(-1.1 < z < -.08)$

$$p(-1.1 < z < -.08) \approx 33.25\%$$



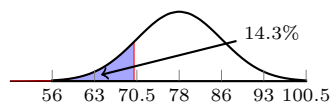


20. A population is normally distributed with a mean 78 and standard deviation 7.5. Find the following probabilities.

a.  $p(x < 70)$

$$z_{70} = \frac{70 - 78}{7.5} = -1.0\bar{6} \approx -1.07$$

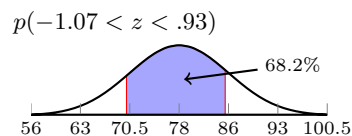
$$p(z < -1.07) \approx 14.23\%$$



b.  $p(70 < x < 85)$

$$z_{70} \approx -1.07 \text{ and } z_{85} = \frac{85 - 78}{7.5} \approx .93$$

$$= p(z < .93) - p(z < -1.07) \approx 68.15\%$$

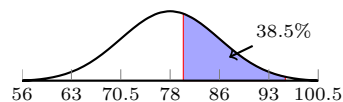


c.  $p(80 < x < 95.5)$

$$z_{80} = \frac{80 - 78}{7.5} \approx .27 \text{ and } z_{95.5} = \frac{95.5 - 78}{7.5} \approx 2.33$$

$$p(.27 < z < 2.33)$$

$$= p(z < 2.33) - p(z < .27) \approx 38.37\%$$

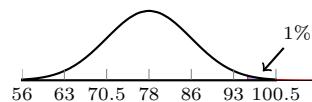


d.  $p(95.5 < x < 108)$

$$z_{95.5} \approx 2.33 \text{ and } z_{108} = \frac{108 - 78}{7.5} = 4$$

$$p(2.33 < z < 4)$$

$$= p(z < 4) - p(z < 2.33) \approx 0.99\%$$

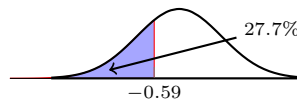


21. Find  $c$  such that each of the following is true.

a.  $p(z < c) = 0.2776$

$$\begin{array}{l} .2776 \rightarrow \text{area} \\ -0.59 \rightarrow \text{z-score} \end{array}$$

$$c = -0.59$$



b.  $p(z > c) = .0351 \leftarrow \text{area in right tail}$

$$\text{Find z-score for } 1 - .0351 = .9649$$

$$\begin{array}{l} .9649 \rightarrow \text{area} \\ 1.81 \rightarrow \text{z-score} \end{array}$$

$$c = 1.81$$



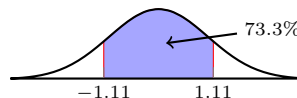
c.  $p(-c < z < c) = .733$

The area is split on either side of the mean,

$$\text{Area} = \frac{.733}{2} = .3665$$

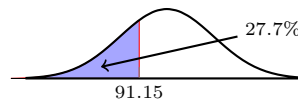
$$\begin{array}{l} .3665 \rightarrow \text{area} \\ \pm 1.11 \rightarrow \text{z-score} \end{array} \text{ (using Appendix F)}$$

$$c = 1.11$$



- d. Let  $\mu = 100$  and  $\sigma = 15$  find  $c$  such that  $p(x < c) = 27.76\%$

$$\begin{aligned} \text{area} & \rightarrow \text{z-score} \\ .2776 & \rightarrow -.59 \\ c = \sigma z + \mu & \Rightarrow c = 15(-.59) + 100 \\ c & = 91.15 \end{aligned}$$

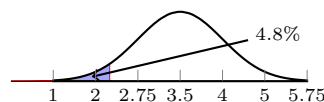


22. The time it takes latex paint to dry is normally distributed with  $\mu = 3.5$  hours and  $\sigma = .75$  hours. Find the probability that drying time will be as follows.

- a. less than 2.25 hours.

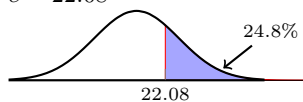
$$z_{2.25} = \frac{2.25 - 3.5}{.75} = -1.6\bar{6} \approx -1.67$$

$$p(x < 2.25) = p(z < -1.67) = 4.75\%$$



- b. The average score on the ACT is 18 with a standard deviation of 6. If a college only wants to accept students with an ACT score at least in the top 25%. What is the minimum score they should accept? What about the top 5%?

**top 25%:**  $\text{area} \rightarrow \approx .67$   
 $\text{z-score}$   
 $c = \sigma z + \mu \Rightarrow c = 6(.68) + 18$   
 $c = 22.08$



**top 5%:**  $\text{area} \rightarrow \approx 1.645$   
 $\text{z-score}$   
 $c = \sigma z + \mu \Rightarrow c = 6(1.65) + 18$   
 $c = 27.9$

